

... Syntax: "\\$Assumptions = b > 0, a > 0;" is incomplete; more input is needed.

In[33]:= **\$Assumptions = b > 0 && a > 0;**
 asunciones

$$\phi = - \frac{GM}{\sqrt{R^2 + \left(a + \sqrt{b^2 + z^2}\right)^2}}$$

Out[34]=
$$- \frac{GM}{\sqrt{R^2 + \left(a + \sqrt{b^2 + z^2}\right)^2}}$$

In[35]:= **d2φdR2 = D[D[φ, R], R]**
 deriva

Out[35]=
$$- \frac{3 GM R^2}{\left(R^2 + \left(a + \sqrt{b^2 + z^2}\right)^2\right)^{5/2}} + \frac{GM}{\left(R^2 + \left(a + \sqrt{b^2 + z^2}\right)^2\right)^{3/2}}$$

In[36]:= **Together[d2φdR2]**
 agrupa

Out[36]=
$$- \frac{GM \left(-a^2 - b^2 + 2 R^2 - z^2 - 2 a \sqrt{b^2 + z^2}\right)}{\left(R^2 + \left(a + \sqrt{b^2 + z^2}\right)^2\right)^{5/2}}$$

In[37]:= **d2φdz2 = D[D[φ, z], z]**
 deriva

Out[37]=
$$- \frac{3 GM z^2 \left(a + \sqrt{b^2 + z^2}\right)^2}{\left(b^2 + z^2\right) \left(R^2 + \left(a + \sqrt{b^2 + z^2}\right)^2\right)^{5/2}} + \frac{GM z^2}{\left(b^2 + z^2\right) \left(R^2 + \left(a + \sqrt{b^2 + z^2}\right)^2\right)^{3/2}} -$$

$$\frac{GM z^2 \left(a + \sqrt{b^2 + z^2}\right)}{\left(b^2 + z^2\right)^{3/2} \left(R^2 + \left(a + \sqrt{b^2 + z^2}\right)^2\right)^{3/2}} + \frac{GM \left(a + \sqrt{b^2 + z^2}\right)}{\sqrt{b^2 + z^2} \left(R^2 + \left(a + \sqrt{b^2 + z^2}\right)^2\right)^{3/2}}$$

In[38]:= **Together[d2φdz2]**
 agrupa

Out[38]=
$$- \left(\left(GM \left(-a^3 b^2 - 3 a b^4 - a b^2 R^2 + a b^2 z^2 + 4 a z^4 - 3 a^2 b^2 \sqrt{b^2 + z^2} - b^4 \sqrt{b^2 + z^2} - \right. \right. \right.$$

$$\left. \left. b^2 R^2 \sqrt{b^2 + z^2} + 2 a^2 z^2 \sqrt{b^2 + z^2} + b^2 z^2 \sqrt{b^2 + z^2} - R^2 z^2 \sqrt{b^2 + z^2} + 2 z^4 \sqrt{b^2 + z^2} \right) \right) /$$

$$\left(\left(b^2 + z^2\right)^{3/2} \left(R^2 + \left(a + \sqrt{b^2 + z^2}\right)^2\right)^{5/2} \right)$$

In[39]:= **dφdR = D[φ, R]**
 ⌞deriva

Out[39]=
$$\frac{GM R}{\left(R^2 + \left(a + \sqrt{b^2 + z^2}\right)^2\right)^{3/2}}$$

\$Assumptions b > 0;

⌞asunciones

x2 = Simplify[Together[(d2φdR2 + 3 dφdR / R) /. {z → 0}]]

⌞simplifica ⌞agrupa

Out[]:=
$$\frac{GM \left(4 a^2 + 8 a b + 4 b^2 + R^2\right)}{\left(a^2 + 2 a b + b^2 + R^2\right)^{5/2}}$$

In[]:= **v2 = Simplify[d2φdz2 /. {z → 0}]**
 ⌞simplifica

Out[]:=
$$\frac{(a + b) GM}{b \left(a^2 + 2 a b + b^2 + R^2\right)^{3/2}}$$

In[]:= **v2 /. {b → 0.1 a}**

Out[]:=
$$\frac{11. GM}{\left(1.21 a^2 + R^2\right)^{3/2}}$$

In[56]:= **φh = $\frac{-G M h}{r + \alpha}$**

Out[56]=
$$-\frac{G M h}{r + \alpha}$$

In[59]:= **\$Assumptions = R > 0;**
 ⌞asunciones

φhcil = φh /. {r → $\sqrt{R^2 + z^2}$ }

Out[60]=
$$-\frac{G M h}{\sqrt{R^2 + z^2} + \alpha}$$

In[61]:= **d2φhdR2 = D[D[φhcil, R], R]**
 ⌞⋯ ⌞deriva

Out[61]=
$$-\frac{2 G M h R^2}{\left(R^2 + z^2\right) \left(\sqrt{R^2 + z^2} + \alpha\right)^3} - \frac{G M h R^2}{\left(R^2 + z^2\right)^{3/2} \left(\sqrt{R^2 + z^2} + \alpha\right)^2} + \frac{G M h}{\sqrt{R^2 + z^2} \left(\sqrt{R^2 + z^2} + \alpha\right)^2}$$

In[62]:= **Simplify[d2φhdR2 /. {z → 0}]**
 ⌞simplifica

Out[62]=
$$-\frac{2 G M h}{\left(R + \alpha\right)^3}$$

In[63]:= **d2φhdz2 = D[D[φhcil, z], z]**
|..|deriva

$$\text{Out[63]} = -\frac{2 G M h z^2}{\left(R^2 + z^2\right) \left(\sqrt{R^2 + z^2} + \alpha\right)^3} - \frac{G M h z^2}{\left(R^2 + z^2\right)^{3/2} \left(\sqrt{R^2 + z^2} + \alpha\right)^2} + \frac{G M h}{\sqrt{R^2 + z^2} \left(\sqrt{R^2 + z^2} + \alpha\right)^2}$$

In[64]:= **Simplify[d2φhdz2 /. {z → 0}]**
|simplifica

$$\text{Out[64]} = \frac{G M h}{R \left(R + \alpha\right)^2}$$

In[67]:= **\$Assumptions = b > 0;**
|asunciones

$$v_c = \sqrt{R \text{Simplify}[D[\phi, R] /. \{z \rightarrow 0\}] + r D[\phi h, r]}$$

$$\text{Out[68]} = \sqrt{\frac{G M R^2}{\left(a^2 + 2 a b + b^2 + R^2\right)^{3/2}} + \frac{G M h r}{\left(r + \alpha\right)^2}}$$

$$\text{Out[55]} = -\frac{G M}{\sqrt{R^2 + \left(a + \sqrt{b^2 + z^2}\right)^2}_h}$$